comes to forming the Grothendieck semigroup or, by abuse of terminology, when we speak of two objects of $\mu\mathrm{VF}[k]$ being isomorphic.

Referee R167 [Typo; new] – Missing verb.
Original source line 1238. In the preceding paragraph, write "that is, *be* an isomorphism" after "should identify the two objects."

Definition 3.8 (RV-categories)**.** The objects of the category $\mathrm{RV}[k]$ are the pairs (U, f) with U a set in RV_∞ and $f : U \longrightarrow \mathrm{RV}^k$ a definable finite-to-one function. Given two such objects (U, f), (V, g), any definable bijection $F : U \longrightarrow V$ is a *morphism* of $\mathrm{RV}[k]$.

By Lemma 2.12, the two categories $\mathrm{VF}[0]$, $\mathrm{RV}[0]$ are actually equivalent, similarly for other such categories.

Notation 3.9. We emphasize that if (U, f) is an object of $\mathrm{RV}[k]$ then $f(U)$ is a subset of RV^k instead of RV_∞^k, while ∞ can occur in any coordinate of U. An object of $\mathrm{RV}[*]$ of the form (U, id) is often just written as U.

More generally, if $f : U \longrightarrow \mathrm{RV}_\infty^k$ is a definable finite-to-one function then (U, f) denote the obvious object of $\mathrm{RV}[\leq k]$. For example, the inclusion $\{\infty\} \longrightarrow \mathrm{RV}_\infty$ gives rise to an object of $\mathrm{RV}[0]$, the inclusion $\{(1, \infty)\} \longrightarrow \mathrm{RV}_\infty^2$ gives rise to an object of $\mathrm{RV}[1]$, and so on. Often f will be a coordinate projection (every object in $\mathrm{RV}[*]$ is isomorphic to an object of this form). In that case, $(U, \mathrm{pr}_{\leq k})$ is simply denoted by $U_{\leq k}$ and its class in $\mathbf{K}_+\,\mathrm{RV}[k]$ by $[U]_{\leq k}$, etc.

Referee R048 [Typo] – Agreement.
Original source line 1252. Change the verb "denote" to "denotes" in the sentence about (U, f).

Definition 3.10 (RES-categories)**.** The category $\mathrm{RES}[k]$ is the full subcategory of $\mathrm{RV}[k]$ such that $(U, f) \in \mathrm{RES}[k]$ if and only if $\mathrm{vrv}(U)$ is finite.

Notation 3.11. Every $\mathrm{RV}[k]$-morphism $F : (U, f) \longrightarrow (V, g)$ induces a definable finite-to-finite correspondence $F^\dagger \subseteq f(U) \times g(V)$, which may also be thought of as a finitary function $f(U) \longrightarrow \mathcal{P}(g(V))$. Recall the operator Jcb_Γ from Notation 2.8. For $u \in U$, we abbreviate $\mathrm{Jcb}_\Gamma\, F^\dagger(f(u), g \circ F(u))$ as $\mathrm{Jcb}_\Gamma\, F^\dagger(u)$

Referee R049 [Typo] – Missing punctuation.
Original source line 1260. Add a full stop after the abbreviation $\mathrm{Jcb}_\Gamma\, F^\dagger(u)$.

Definition 3.12 (RV- and RES-categories with Γ-volume forms)**.** An object of the category $\mu\mathrm{RV}[k]$ is a definable triple (U, f, ω), where (U, f) is an object of $\mathrm{RV}[k]$ and $\omega : U \longrightarrow \Gamma$ is a function, which is understood as a *definable Γ-volume form* on (U, f). A *morphism* between two such objects (U, f, ω), (V, g, σ) is an $\mathrm{RV}[k]$-morphism $F : (U, f) \longrightarrow (V, g)$ such that for every $u \in U$,

$$\omega(u) = (\sigma \circ F)(u) + \mathrm{Jcb}_\Gamma\, F^\dagger(u). \tag{3.2}$$

The category $\mu\mathrm{RES}[k]$ is the obvious full subcategory of $\mu\mathrm{RV}[k]$.

Definition 3.13 (Γ-categories)**.** The objects of the category $\Gamma[k]$ are the finite disjoint unions of definable subsets of Γ^k. A definable bijection between two such objects is a morphism of $\Gamma[k]$ if and only if it is the vrv-contraction of a definable bijection.

The category $\Gamma^{\mathrm{fin}}[k]$ is the full subcategory of $\Gamma[k]$ such that $I \in \Gamma^{\mathrm{fin}}[k]$ if and only if I is finite.

Remark 3.14. If $\gamma \in \Gamma$ is definable then it is in the divisible hull of $\Gamma(\mathbb{S})$. This does not mean that the definable set $\gamma^\sharp \subseteq \mathrm{RV}$ contains a definable point unless $\gamma \in \Gamma(\mathbb{S})$.

Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms.
Formalization question C001: the coefficient embedding. The corrected Proposition 3.18, both of its positive kernel presentations, and the degree-zero category comparisons have been checked. The remaining implication for this square is the following, with the original ground and every finite disjoint-union tag retained:

$$I^\sharp \simeq_{\mathbb{S}} J^\sharp \quad \Longrightarrow \quad I \simeq_{\Gamma[n]} J.$$

More explicitly, given a definable bijection $F : I^\sharp \longrightarrow J^\sharp$, we must construct a definable bijection $f : I \longrightarrow J$ and some definable bijection $F' : I^\sharp \longrightarrow J^\sharp$ with $\mathrm{vrv} \circ F' = f \circ \mathrm{vrv}$. The replacement F' need not equal F. The case $n = 0$, and the case where each Gamma domain is a singleton, are proved. The unresolved case includes finite unions of value fibres whose Gamma values have no definable RV representatives.
Author input requested: what argument should supply this replacement over the original ground, in particular in that finite-fibre case? The tensor quotient theorem does not itself prove this implication. This is a proof question, not a proposed change to the statement, and no counterexample to the coefficient embedding is being asserted.

The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$